

MODULE 25

AI Ethics and the Future of AI

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

AI Ethics and the Future of AI

Why AI Ethics Matters

AI systems make decisions that affect people's lives:
Credit scoring, hiring, medical diagnosis, criminal justice

Key ethical concerns:

Bias and fairness

Privacy and surveillance

Transparency and explainability

Accountability

Job displacement

Environmental impact

AI safety and alignment

Bias in AI

Types of bias:

Data bias: training data reflects historical prejudices

Algorithm bias: model amplifies existing biases

Evaluation bias: metrics don't reflect real-world performance

Examples:

Facial recognition worse for dark skin

Hiring algorithms discriminating by gender

Credit scoring penalising zip codes

Check for bias

```
from aif360.metrics import BinaryLabelDatasetMetric
metric = BinaryLabelDatasetMetric(dataset, privileged_groups, unprivileged_groups)
print('Disparate Impact:', metric.disparate_impact())
```

Explainable AI XAI

Black-box models: we don't know why they decide.

Explainability: understand model decisions.

```
import shap
```

```
# SHAP values for feature importance
```

```
explainer = shap.TreeExplainer(model)
```

```
shap_values = explainer.shap_values(X_test)
```

```
# Summary plot: overall feature importance
```

```
shap.summary_plot(shap_values, X_test, feature_names=feature_names)
```

```
# Force plot: single prediction explanation
```

```
shap.force_plot(explainer.expected_value, shap_values[0], X_test[0])
```

```
# LIME for local explanations
import lime.lime_tabular
explainer = lime.lime_tabular.LimeTabularExplainer(X_train)
exp = explainer.explain_instance(X_test[0], model.predict_proba)
```

AI Safety and Alignment

AI Alignment: ensuring AI systems pursue goals aligned with human values.

Key risks:

Misaligned objectives: optimising wrong metrics

Distribution shift: real world differs from training

Adversarial attacks: carefully crafted inputs fool AI

Automation bias: over-relying on AI decisions

Safety techniques:

Reinforcement learning from human feedback (RLHF)

Constitutional AI (Anthropic)

Robustness testing

Formal verification

Human-in-the-loop systems

Future of AI

Near term (2025-2030):

Multimodal AI (text + image + audio + video)

AI agents: autonomous task completion

Personalised AI assistants

AI-accelerated drug discovery

Autonomous vehicles Level 4-5

Long term:

Artificial General Intelligence (AGI): debated timeline

Human-AI collaboration

Scientific breakthroughs (protein folding, materials)

Career advice:

Stay current: read papers, follow conferences

Build projects: learn by doing

Collaborate: join AI communities

Specialise: pick an application domain